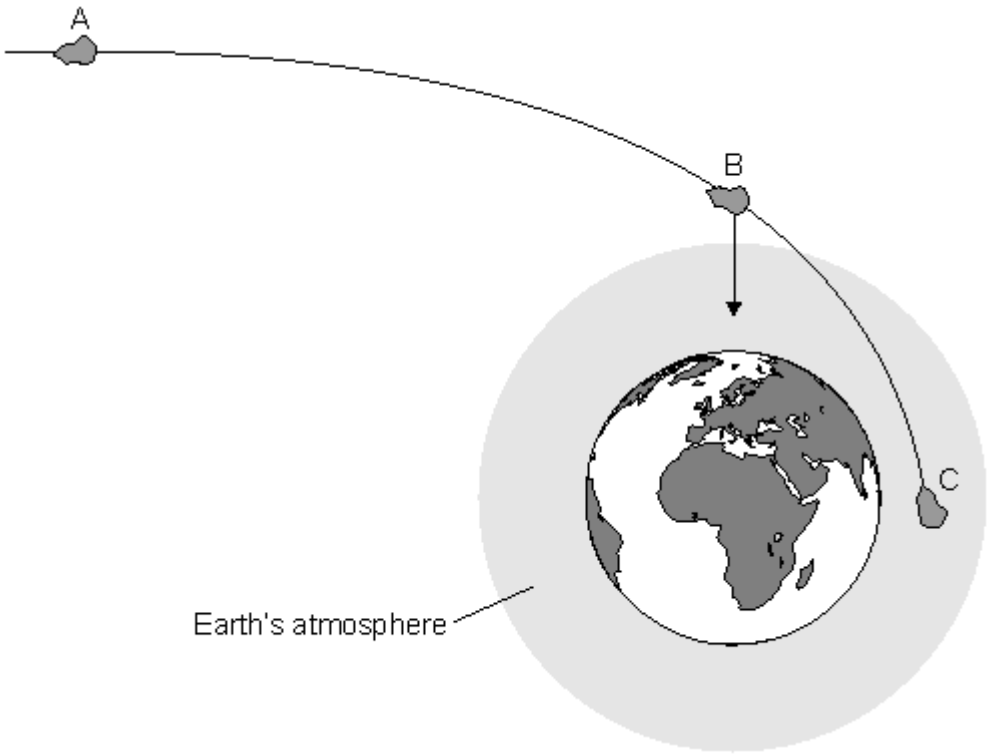
Q1. The diagram below shows the path of a meteor as it gets closer to the Earth.
The meteor is shown in three positions: A, B and C.



not to scale

(a) The path of the meteor is affected by the Earth's gravity.
The arrow shows the direction of the force due to gravity acting on the meteor at B.

(i) **On the diagram** draw an arrow to show the direction of the force of gravity on the meteor at A.
Use a ruler.

1 mark

(ii) **On the diagram** draw an arrow to show the direction of the force of gravity on the meteor at C.
Use a ruler.

1 mark

(iii) How does the force of gravity on the meteor change as it travels from A to C?

.....

1 mark

(b) What happens to the speed of the meteor as it travels from A to B?

.....

1 mark

(c) When the meteor enters the Earth's atmosphere, three forces act on the meteor. Gravity and upthrust are two of these forces.

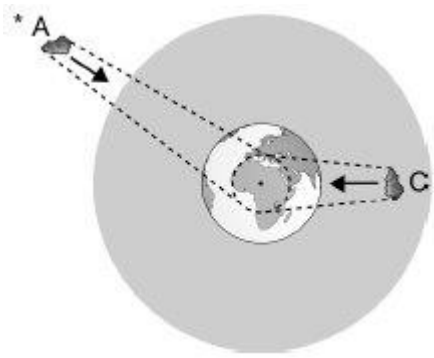
Give the name of the **other** force.

.....

1 mark
maximum 5 marks

Q1.

(a) (i)(ii) •



accept arrows drawn within the tolerance of half the diameter of the Earth shown by the dashed lines

*do **not** accept arrows outside tolerances shown*
award one mark for an arrow from the meteor in position A
award one mark for an arrow from the meteor in position C

2 (L5)

(iii) any **one** from

- it increases

  accept 'it gets stronger'
  accept 'it attracts more'

- it changes direction

  'it curves round' is insufficient

1 (L5)

(b) any **one** from

- increases **or** goes up

  accept 'it accelerates'

- it gets faster

1 (L5)

(c) any **one** from

- air resistance

  accept 'friction'
  *do **not** accept 'wind resistance'*

- drag

  accept 'lift'
  'resistance' is insufficient
  *do **not** accept 'weight' **or** 'upthrust' **or** 'gravity'*

1 (L6)

[5]